Comprehensive Mobility Dashboard for the Trucking Sector

IUCRC Phase 1, Year 1

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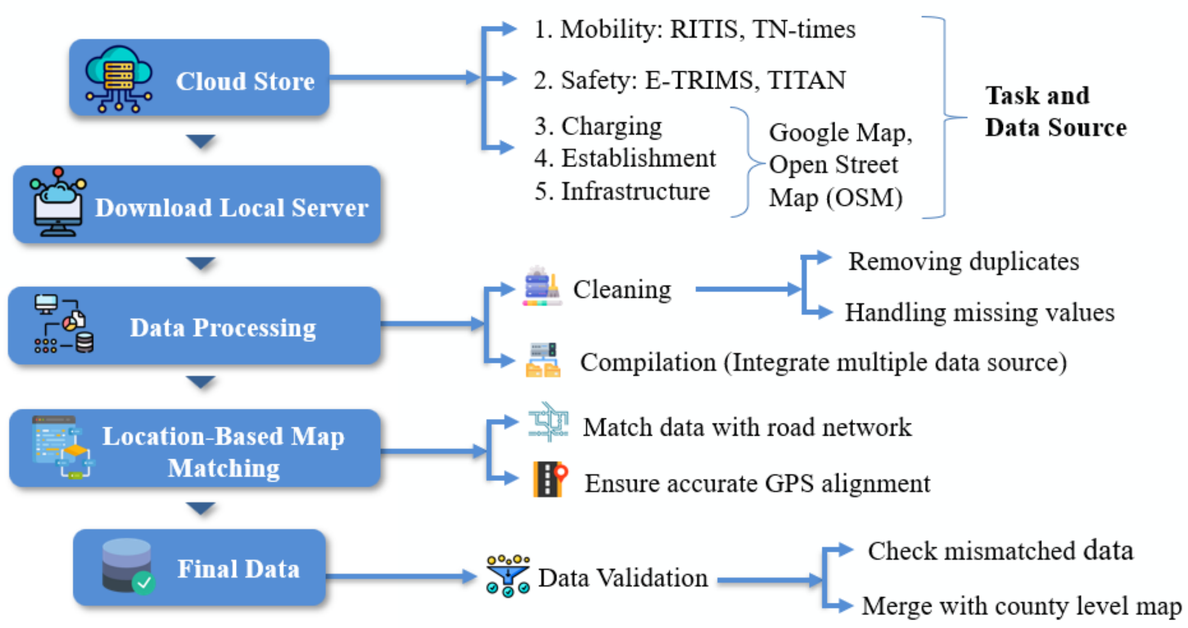
Efficient freight transport is key to economic sustainability, but truck crashes and congestion hinder mobility and safety. This study identifies high-risk areas on interstates and highways, highlighting speed variations and intersection-related crashes. Strategies like adaptive signals, truck-priority lanes, and dynamic routing can enhance mobility and reduce crash risks, improving freight efficiency and road safety.

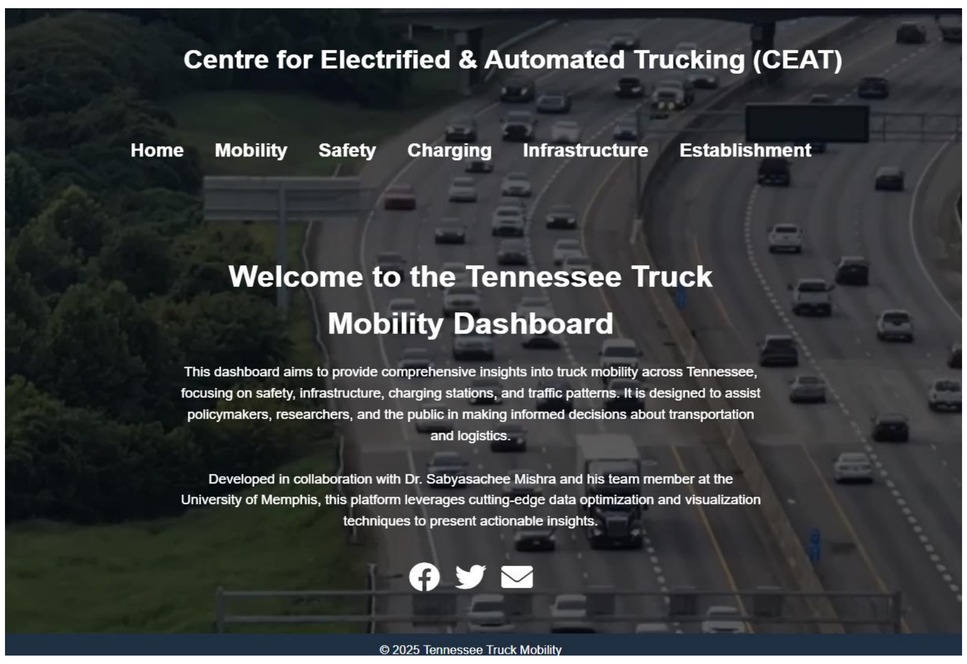
Objectives

- Multi source data collection, integration and trend analysis.
- Build data driven real-time models for traffic monitoring.
- Develop interactive dashboard and software.
- Evaluate existing road networks and generate report.

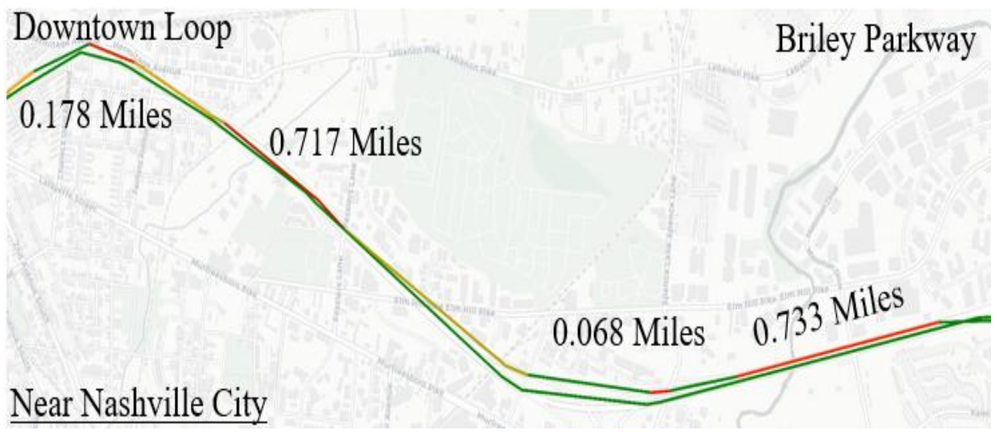
Methodology



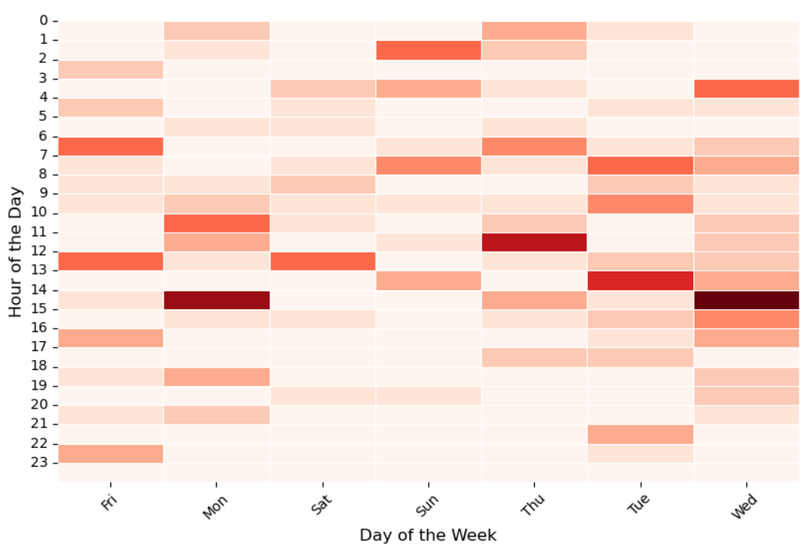
Dashboard



Congestion

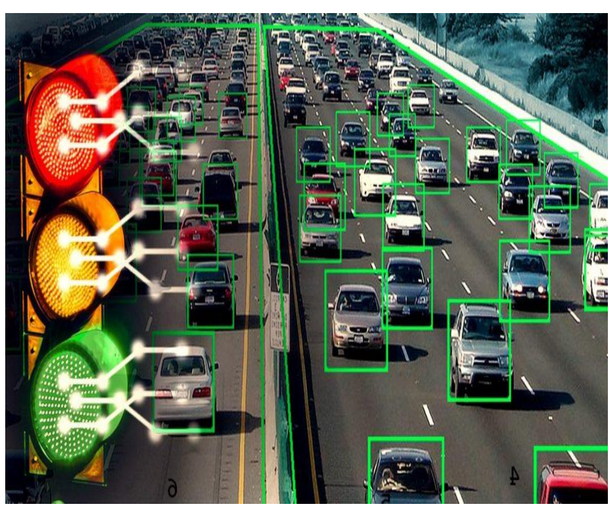


Speed Heatmap



Spatiotemporal Truck Crash

EV Charging Stations for Truck



Data Driven Method